

Hardware Architecture & Schematic Design

Low-Cost Smart Pen System for Parkinson's Disease Tremor Screening
ESP32-S3 SuperMini · MPU-6050 · Tri-FSR Array · Option 2 Boot-Safe RC Decoupling
Version 2.0 · June 2026

! **DISCLAIMER:** This system is a non-diagnostic screening and monitoring tool. It is not cleared as a medical device. All output must be interpreted alongside professional clinical assessment using validated scales (e.g. MDS-UPDRS).

1. System Overview

The hardware architecture balances high-fidelity multi-channel data acquisition, edge-computing performance, and strict spatial constraints imposed by the pen form factor. The system comprises three primary sub-circuits: the Central Processing Subsystem (ESP32-S3 SuperMini), the Sensor Acquisition Network (MPU-6050 + Tri-FSR array), and the Power Management Subsystem (LiPo + LDO). All subsystems share a single stable 3.3 V regulated rail to prevent analog FSR readings from being corrupted by battery discharge variation.

1.1 Signal Pipeline

Sensor Acquisition → ADC1 sampling @ 1 kHz → DC offset removal
→ High-pass filter (1 Hz) → Band-pass filter (3.5–6.5 Hz)
→ 1024-point FFT (on-device) → Feature extraction
→ Native BLE transmission @ 1 Hz → Mobile application

1.2 Component Summary

Subsystem	Component	Key Details
Central processing	ESP32-S3 SuperMini	Dual-core LX7 @ 240 MHz · native BLE 5 · ADC1 channels · onboard LDO
Kinematic sensing	MPU-6050 MEMS IMU	I2C digital · 4–6 Hz tremor range · 100 nF VCC decoupling · AD0 → GND (0x68)
Axial force sensing	FSR-400 (tip)	7.5 mm active area · plunger-isolated · Option 2 RC circuit on GPIO 1
Grip force sensing	FSR-402 (barrel A)	44 mm active area · fingerprint zone · Option 2 RC

Subsystem	Component	Key Details
A		circuit on GPIO 2
Grip force sensing B	FSR-402 (barrel B)	44 mm active area · fingerprint zone · Option 2 RC circuit on GPIO 3
Power	LiPo 3.7 V · 500 mAh	4–6 hr runtime · onboard LDO → flat 3.3 V rail for ADC stability

2. Central Processing Subsystem — ESP32-S3 SuperMini

The ESP32-S3 SuperMini is an ultra-compact development board housing the dual-core Xtensa 32-bit LX7 microprocessor. It was selected over standard ARM Cortex-M or 8-bit microcontrollers for three reasons: sufficient computational headroom for 1 kHz multi-channel sampling with concurrent FFT processing, native BLE 5 eliminating the need for an external UART module, and an onboard LDO providing the stable 3.3 V rail essential for reliable analog FSR readings.

2.1 Key Specifications

Parameter	Value
MCU	ESP32-S3 (Xtensa dual-core LX7)
Clock speed	Up to 240 MHz (dual-core)
Operating voltage	3.3 V logic (regulated by onboard LDO)
Wireless	Native BLE 5 · 2.4 GHz onboard PCB antenna · no external module required
ADC resolution	12-bit nominal · ~11-bit effective with noise
ADC channels used	ADC1 only (GPIO 1, 2, 3, 4) · ADC2 avoided — incompatible with active BLE
I2C	GPIO 8 (SDA) · GPIO 9 (SCL)
Flash / PSRAM	8 MB flash · 2 MB PSRAM (model-dependent)

2.2 GPIO Pin Assignment

Pins are partitioned to separate the high-speed I2C digital bus from the sensitive analog ADC channels, preventing crosstalk-induced noise on FSR traces. ADC2 channels are avoided entirely as they share internal hardware paths with the BLE radio stack and produce unreliable readings during active wireless transmission.

GPIO	Function	Interface	Notes
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GPIO	Function	Interface	Notes
GPIO 8	I2C SDA	MPU-6050 data	4.7 kΩ pull-up to 3.3 V required
GPIO 9	I2C SCL	MPU-6050 clock	4.7 kΩ pull-up to 3.3 V required
GPIO 1	ADC1_CH1	FSR-400 tip sensor	Strapping pin — Option 2 RC mitigation applied
GPIO 2	ADC1_CH2	FSR-402 grip A	Strapping pin — Option 2 RC mitigation applied
GPIO 3	ADC1_CH3	FSR-402 grip B	Strapping pin — Option 2 RC mitigation applied

! STRAPPING PIN WARNING: GPIO 0, 1, 2, and 3 are ESP32-S3 strapping pins sampled during boot to determine startup mode. A fixed voltage divider connected directly to these pins can hold an incorrect voltage level during boot, causing boot loops or preventing firmware upload. Option 2 RC decoupling (Section 4) is mandatory on all three FSR channels to mitigate this risk.

3. Sensor Acquisition Network

The sensor network uses two distinct electrical interfaces: a digital I2C bus for the MPU-6050 IMU and a passive analog voltage divider front-end for each of the three FSRs. The two interface types are electrically isolated through GPIO partitioning to prevent digital switching noise from coupling into analog FSR traces.

3.1 Kinematic Vibration Sensor — MPU-6050

The MPU-6050 is a MEMS-based 6-axis IMU (3-axis accelerometer + 3-axis gyroscope) mounted internally near the writing tip to maximise sensitivity to vertical tremor oscillations in the 4–6 Hz Parkinsonian frequency range. It communicates via I2C at up to 400 kHz.

Electrical Connections

MPU-6050 Pin	Connects To	Notes
VDD (pin 13)	3.3 V rail	Add 100 nF ceramic decoupling cap physically close to this pin
GND (pin 18)	GND	System ground
SDA (pin 23)	GPIO 8	4.7 kΩ pull-up to 3.3 V · verify breakout board does not already include pull-ups
SCL (pin 24)	GPIO 9	4.7 kΩ pull-up to 3.3 V · verify breakout board does not already include pull-ups

MPU-6050 Pin	Connects To	Notes
AD0 (pin 11)	GND	Sets I2C address to 0x68 · must not be left floating
INT, FSYNC	No Connect (NC)	Leave unconnected · pull to GND via 10 kΩ if noise is observed
VLOGIC (pin 8)	3.3 V rail	Tie to 3.3 V on variants that expose this pin
CLKIN (pin 2)	NC	Internal oscillator used
AUX_DA, AUX_CL	NC	Secondary I2C bus not used in this design
RESV pins	NC	Reserved — do not connect

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NOTE: The 100 nF VCC decoupling capacitor on MPU-6050 pin 13 must be placed within 1 mm of the pin pad on a PCB layout. At greater distances it becomes ineffective at suppressing BLE radio-frequency interference on the 3.3 V rail.

3.2 Tri-FSR Analog Front-End

Three Force Sensitive Resistors capture pressure dynamics without requiring bulky operational amplifiers or load cells. Each FSR is configured in a passive voltage divider circuit with identical component values, forming three independent ADC1 input channels.

Sensor Selection Rationale

Sensor	Active Area	GPIO	Placement	Selection Rationale
FSR-400	7.5 mm diameter	GPIO 1 (ADC1_CH1)	Inside barrel — ink cartridge distal end	Small area fits the narrow tip channel. A micro-plunger transfers only axial (downward) force, mechanically isolating it from lateral grip pressure.
FSR-402 A	44 mm diameter	GPIO 2 (ADC1_CH2)	Lower barrel — primary finger contact	Large area covers the full fingerprint region regardless of individual grip style. Captures both sustained grip rigidity and high-frequency micro-tremor oscillations.
FSR-402 B	44 mm diameter	GPIO 3 (ADC1_CH3)	Lower barrel — secondary finger contact	Eliminates blind spots when grip contact shifts between users. Two sensors working in tandem provide redundancy and capture pill-rolling micro-tremors regardless of finger positioning.

Base Voltage Divider Circuit (identical for all three FSRs)

Each FSR terminal 1 connects to 3.3 V. Terminal 2 connects to Node A — a junction point shared with a 10 kΩ pull-down resistor to GND and a 100 nF capacitor in parallel with the pull-down. As

applied force increases, FSR resistance drops from >10 MΩ (unloaded) toward ~10 kΩ (firmly pressed), raising the Node A voltage proportionally from near 0 V toward 3.3 V.

RC Low-Pass Hardware Filter

The 100 nF capacitor in parallel with the 10 kΩ pull-down resistor forms a passive RC low-pass filter:

$$f_c = 1 / (2\pi \times 10,000 \, \Omega \times 100 \times 10^{-9} \, \text{F}) \approx 159 \, \text{Hz}$$

This filter passes the 4–6 Hz biological tremor signal with negligible attenuation while rejecting 50/60 Hz mains interference and high-frequency ADC switching noise.

4. Option 2 Boot-Safe RC Decoupling — All Three FSR Channels

Because GPIO 1, 2, and 3 are ESP32-S3 strapping pins, a direct voltage divider connection risks holding these pins at an incorrect voltage level during boot, causing boot loops or preventing firmware upload. Option 2 adds a 100 kΩ series resistor and a 10 μF electrolytic boot-delay capacitor between Node A (the FSR divider output) and the GPIO pin (Node B).

4.1 How Option 2 Works

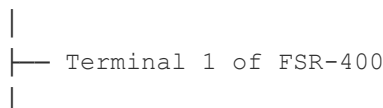
- **During boot (0–~2 ms)::** The ESP32-S3 samples strapping pins to determine startup mode. Node B (the GPIO pin) is held near 0 V by the 10 μF capacitor, which has not yet charged. The ESP32 reads the correct strapping value.
- **RC time constant::** $\tau = 100 \, \text{k}\Omega \times 10 \, \mu\text{F} = 1 \, \text{second}$. Node B rises slowly toward Node A over approximately 1 second after power-on.
- **During normal operation (after ~3 s)::** Node B has settled and tracks the FSR pressure signal normally. Firmware enforces a 3-second startup delay before ADC sampling begins.
- **100 kΩ series resistor function::** Makes the FSR divider a high-impedance source from the GPIO's perspective during boot. The ESP32-S3 internal strapping logic easily overrides it.

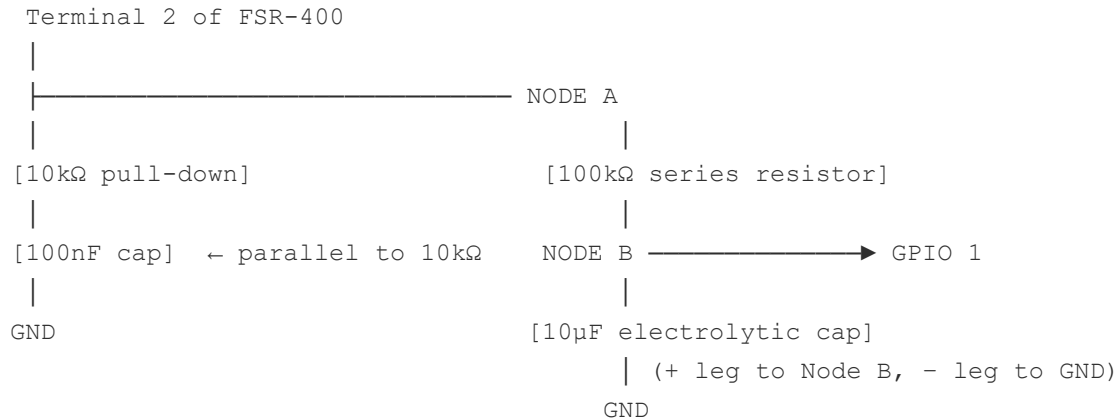
4.2 Complete Circuit Per FSR Channel

The following circuit applies identically to all three FSR channels. Only the FSR type and GPIO pin differ.

FSR-400 (Tip Sensor) → GPIO 1 (ADC1_CH1)

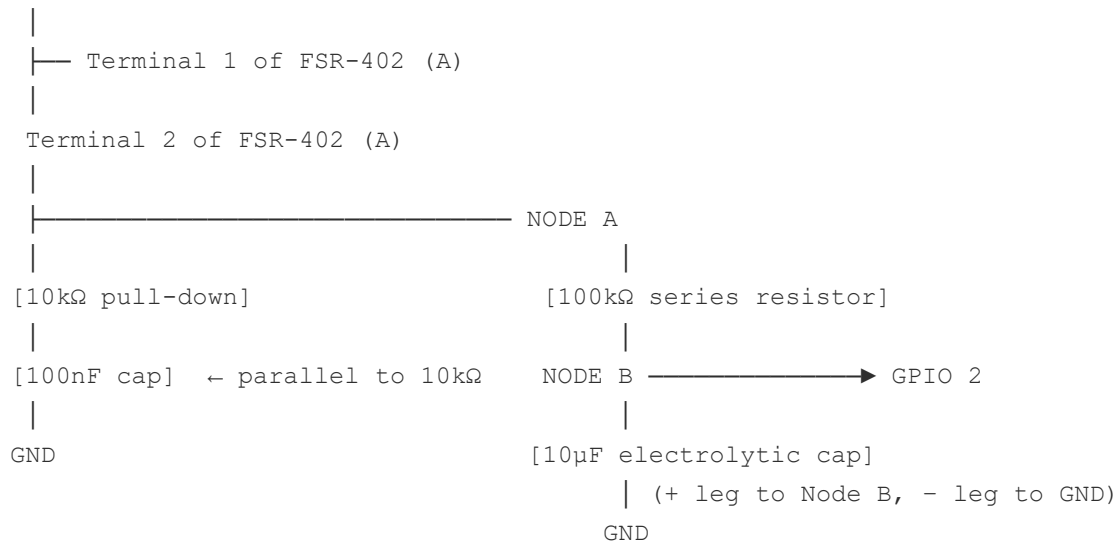
3.3V





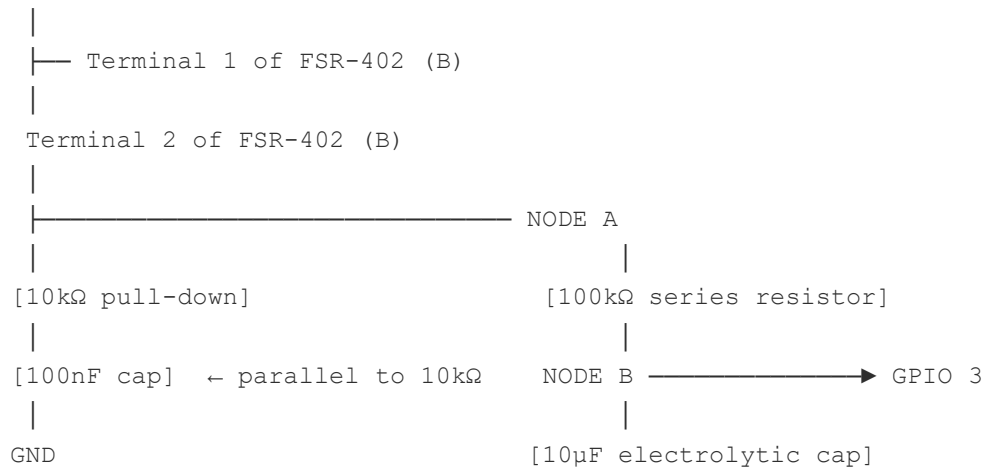
FSR-402 Grip A → GPIO 2 (ADC1_CH2)

3.3V



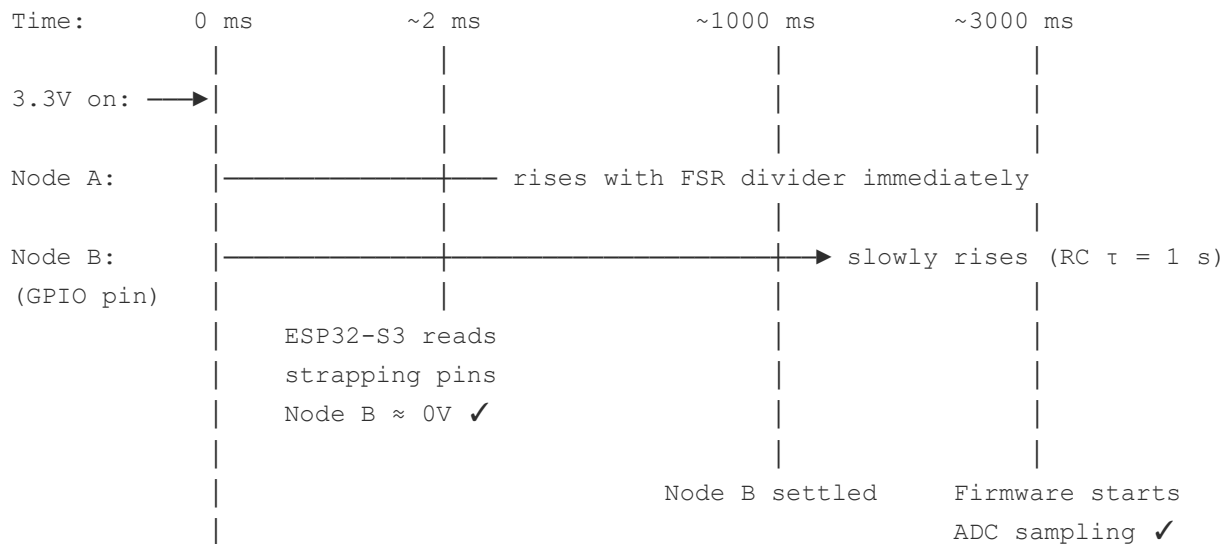
FSR-402 Grip B → GPIO 3 (ADC1_CH3)

3.3V



| (+ leg to Node B, - leg to GND)
GND

4.3 Boot Timing Diagram



! POLARITY WARNING: The 10 μF capacitors are electrolytic and polarised. The positive leg (+) must connect to Node B (the GPIO side). The negative leg (-) must connect to GND. Reversed installation will damage the capacitor and may damage the ESP32-S3.

i FIRMWARE NOTE: Add a 3-second startup delay in firmware before beginning ADC sampling. This allows Node B to fully settle after power-on and prevents false readings during the RC charging period.

5. Power Management Subsystem

The pen operates entirely untethered. Stability on the 3.3 V rail is critical: any voltage fluctuation corrupts analog FSR readings by shifting the baseline voltage of all three voltage dividers simultaneously, introducing false tremor artefacts that cannot be distinguished from real signal.

Parameter	Value
Connector	JST-PH 2-pin (positive + negative only)
Battery chemistry	Lithium-polymer (LiPo)
Nominal voltage	3.7 V
Capacity	500 mAh

Parameter	Value
Voltage range	4.2 V (fully charged) → 3.2 V (depleted)
Input connection	VBAT / 5V pad on ESP32-S3 SuperMini
Regulated output	3.3 V (flat) via onboard Low-Dropout (LDO) linear regulator
Current draw	~80–120 mA (active BLE + dual-core DSP)
Estimated runtime	4–6 hours continuous — exceeds standard clinical session duration

JST CONNECTOR NOTE: The schematic uses a 2-pin JST-PH connector (CN1) for the LiPo battery — positive (+) and negative (–) only. A 4-pin JST-PH would imply a balance connector used in multi-cell packs. This design uses a single-cell LiPo and requires only the 2-pin variant.

6. Complete Schematic Pinout & Connection Matrix

This table defines every hardware connection in the system. It reflects all corrections from the schematic review: corrected MCU label (ESP32-S3 not C3), AD0 explicitly grounded, VLOGIC tied to 3.3 V, decoupling cap on MPU-6050 VDD, Option 2 RC decoupling on all FSR channels, and 2-pin JST-PH connector.

Component	Component Pin	ESP32-S3 Pin / Net	Value / Notes
LiPo battery	(+) positive	VBAT / 5V pad	JST-PH 2-pin connector · raw 3.7–4.2 V input
LiPo battery	(–) negative	GND	System ground
MPU-6050	VDD (pin 13)	3.3 V	100 nF ceramic decoupling cap to GND · place within 1 mm
MPU-6050	GND (pin 18)	GND	System ground
MPU-6050	SDA (pin 23)	GPIO 8	I2C data · 4.7 kΩ pull-up to 3.3 V (verify breakout)
MPU-6050	SCL (pin 24)	GPIO 9	I2C clock · 4.7 kΩ pull-up to 3.3 V (verify breakout)
MPU-6050	AD0 (pin 11)	GND	Sets I2C address 0x68 · must not float
MPU-6050	VLOGIC (pin 8)	3.3 V	Tie to 3.3 V · do not leave floating
MPU-6050	INT, FSYNC	NC	Optional: 10 kΩ to GND if EMI observed
MPU-6050	RESV, AUX_DA,	NC	Reserved / unused · do not

Component	Component Pin	ESP32-S3 Pin / Net	Value / Notes
	AUX_CL, CLKIN		connect
FSR-400 (tip)	Terminal 1	3.3 V	Active power
FSR-400 (tip)	Terminal 2	Node A	Junction: 10 kΩ to GND 100 nF to GND
R (series)	Node A → Node B	100 kΩ	Boot-safe series resistor (Option 2)
C (boot delay)	Node B → GND	10 μF electrolytic	Boot delay cap · (+) to Node B · (-) to GND
FSR-400 (tip)	Node B	GPIO 1 (ADC1_CH1)	Final GPIO connection after RC network
FSR-402 grip A	Terminal 1	3.3 V	Active power
FSR-402 grip A	Terminal 2	Node A	Junction: 10 kΩ to GND 100 nF to GND
R (series)	Node A → Node B	100 kΩ	Boot-safe series resistor (Option 2)
C (boot delay)	Node B → GND	10 μF electrolytic	Boot delay cap · (+) to Node B · (-) to GND
FSR-402 grip A	Node B	GPIO 2 (ADC1_CH2)	Final GPIO connection after RC network
FSR-402 grip B	Terminal 1	3.3 V	Active power
FSR-402 grip B	Terminal 2	Node A	Junction: 10 kΩ to GND 100 nF to GND
R (series)	Node A → Node B	100 kΩ	Boot-safe series resistor (Option 2)
C (boot delay)	Node B → GND	10 μF electrolytic	Boot delay cap · (+) to Node B · (-) to GND
FSR-402 grip B	Node B	GPIO 3 (ADC1_CH3)	Final GPIO connection after RC network

7. PCB Layout Considerations

When transitioning from the breadboard prototype to a custom PCB fitted inside the pen chassis, adhere strictly to the following constraints.

7.1 Antenna Keep-Out Zone

The ESP32-S3 SuperMini PCB antenna must not be covered by any copper plane, metallic housing, or conductive shielding. Define a keep-out zone extending at least 3 mm beyond the antenna traces in all directions. No metallic pen barrel components should sit directly above or below the antenna region.

7.2 Analog / Digital Trace Isolation

Route I2C digital traces (SDA, SCL on GPIO 8, 9) as far as possible from the three analog ADC traces (GPIO 1, 2, 3). High-speed I2C clock edges switching at 100–400 kHz can capacitively couple into adjacent FSR traces, producing periodic noise artefacts that mimic micro-tremors. Use a ground-fill guard trace between the digital and analog routing zones where board area permits.

7.3 Decoupling Capacitor Placement

Capacitor	Target	Maximum Distance	Note
100 nF (×3, RC filter)	FSR pull-down resistor	2 mm from ADC pin pad	One per FSR channel — Node A
10 µF (×3, boot delay)	GPIO pin (Node B)	2 mm from GPIO pin pad	Electrolytic — observe polarity
100 nF (MPU-6050 decoupling)	MPU-6050 VDD pin 13	1 mm from VDD pin	Critical — BLE RF suppression

7.4 Ground Plane

Use a solid copper ground pour across the entire PCB bottom layer. This provides a low-impedance return path for analog signals, reduces susceptibility to external EMI, and acts as a passive heat sink for the ESP32-S3 during sustained BLE transmission.

7.5 FSR Mechanical Interface

- **FSR-400 tip:** Requires a dedicated micro-plunger channel machined into the pen barrel. The channel must constrain lateral movement so only axial (downward) force from the ink cartridge reaches the sensor active area.
- **FSR-402 grip sensors:** Must be conformally adhered to the curved barrel surface using a thin compliant adhesive layer (e.g. silicone transfer tape) to ensure uniform contact pressure across the 44 mm active area without creating rigid pressure hot-spots.

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LIFECYCLE NOTE: Formal lifecycle testing of the plunger mechanism under repeated ballpoint writing impact is required before clinical deployment. High-frequency strokes may degrade the plunger channel geometry over time, altering axial force transfer characteristics and introducing drift in the FSR-400 baseline.

8. Signal Integrity & Noise Budget

The ESP32-S3 ADC achieves approximately 11-bit effective resolution in practice. The following layered noise mitigation strategy maintains signal quality for 4–6 Hz tremor detection across all three FSR channels.

Technique	Stage	Effect
RC hardware LPF ($f_c \approx 159$ Hz)	Analog front-end (Node A)	Reduces broadband noise floor before ADC sampling
100 k Ω series + 10 μ F boot delay	Boot protection (Node B)	Isolates GPIO from divider during strapping pin sampling
Software oversampling (4 $\times$ average)	Firmware	Recovers ~ 0.5 effective ADC bits
DC offset removal	DSP chain	Eliminates slow drift from FSR viscoelastic creep
Band-pass filter 3.5–6.5 Hz	DSP chain	Isolates Parkinsonian tremor band in software
ADC1-only routing	Hardware architecture	ADC2 shares BLE radio paths — avoided entirely
Digital/analog GPIO separation	PCB layout	Prevents I2C clock coupling into FSR ADC traces

8.1 Frequency Domain Budget

Signal of interest:	4–6 Hz	(Parkinsonian tremor band)
RC filter cutoff:	~ 159 Hz	(Node A hardware LPF)
High-pass filter (SW):	1 Hz	(DC offset removal)
Band-pass filter (SW):	3.5–6.5 Hz	(DSP tremor isolation)
FFT window:	1024 samples @ 1 kHz = 1.024 s	
FFT frequency resolution:	~ 0.98 Hz/bin	
ADC sample rate:	1 kHz per channel	

9. Complete Component Bill of Materials

Ref	Component	Value / Part No.	Qty	Purpose
U1	ESP32-S3 SuperMini	ESP32-S3 (dual-core LX7)	1	MCU · BLE · ADC · LDO
U2	IMU	MPU-6050	1	6-axis MEMS IMU · I2C

Ref	Component	Value / Part No.	Qty	Purpose
U3	Tip force sensor	FSR-400	1	Axial writing force · 7.5 mm
U4, U5	Grip force sensors	FSR-402	2	Fingerprint grip · 44 mm
BT1	LiPo battery	3.7 V 500 mAh	1	Main power · JST-PH 2-pin
CN1	Battery connector	JST-PH 2-pin	1	LiPo interface · 2-pin only
H1	Tip FSR header	2-pin male header	1	FSR-400 connection
H4, H5	Grip FSR headers	2-pin male header	2	FSR-402 A and B connections
R1, R5, R8	FSR pull-down resistors	10 kΩ	3	Voltage divider lower leg (Node A)
R3, R7, R9	Boot-safe series resistors	100 kΩ	3	Option 2 · Node A to Node B isolation
R2, R6	I2C pull-up resistors	4.7 kΩ	2	SDA + SCL pull-ups (if not on breakout)
C1, C4, C7	FSR RC filter capacitors	100 nF ceramic	3	Hardware LPF · parallel to 10 kΩ pull-down
C3, C6, C8	Boot delay capacitors	10 μF electrolytic	3	Option 2 · Node B to GND · observe polarity
C2	MPU-6050 decoupling cap	100 nF ceramic	1	VDD pin 13 · place within 1 mm
R4	Floating — delete	100 kΩ	0	REMOVE — unconnected in schematic v1

10. Schematic Review — Corrections Applied (v1 → v2)

The following issues were identified in schematic v1 and corrected in this v2 document.

#	Severity	Issue	Correction Applied
1	Critical	MCU labeled 'ESP32-C3 SuperMini' — wrong chip	Corrected to ESP32-S3 SuperMini throughout
2	Critical	GPIO 1, 2, 3 are strapping pins — direct FSR divider connection risks boot loops	Option 2 RC decoupling applied: 100 kΩ series + 10 μF boot delay cap per channel
3	Critical	MPU-6050 VDD decoupling capacitor missing	100 nF ceramic cap added to VDD pin 13
4	Warning	R4 (100 kΩ) floating — not connected	R4 removed from BOM and schematic
5	Warning	AD0 pin shown as NC — must not float	AD0 explicitly tied to GND · I2C address fixed at 0x68

#	Severity	Issue	Correction Applied
6	Warning	VLOGIC pin disposition not specified	VLOGIC tied to 3.3 V rail
7	Warning	CN1 shown as 4-pin connector — incorrect for single-cell LiPo	Changed to 2-pin JST-PH (positive + negative only)
8	Info	I2C net labels (I2C_SDA, I2C_SCL) ambiguity on ESP32 block	Confirmed: I2C_SDA → GPIO 8 · I2C_SCL → GPIO 9 · documented in pinout table

11. Ethical & Data Safety Considerations

- **BLE encryption:** All BLE transmission must use AES-128 encryption before any clinical deployment to prevent interception of patient kinematic data.
- **On-device storage:** The companion mobile application must use encrypted on-device storage (SQLCipher on Android · iOS Data Protection class NSFileProtectionComplete).
- **Cloud synchronisation:** Any cloud sync of feature vectors must comply with HIPAA (US) or GDPR (EU), including explicit informed patient consent and data minimisation.
- **Clinical representation:** The device must never be represented to patients or clinicians as a diagnostic instrument. Results must always accompany professional clinical assessment.
- **Data retention:** Feature vectors and raw session logs must be subject to a defined retention policy agreed with the institution's data protection officer before deployment.

12. Revision History

Version	Date	Changes
v1.0	June 2026	Initial hardware architecture document
v2.0	June 2026	Corrected MCU label (C3→S3) · Added Option 2 RC boot-safe decoupling on all FSR channels · Added MPU-6050 VDD decoupling cap · Grounded AD0 · Tied VLOGIC to 3.3 V · Removed floating R4 · Corrected CN1 to 2-pin JST-PH · Full schematic corrections table added

For signal processing and firmware documentation see [SIGNAL_PROCESSING.md](#) · For mobile application see [MOBILE_APP.md](#)